

MAC-106 (DBMS): Assignment 1

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Assignment Solutions

- Q1. Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period of time, and has an associated due date, and the date when the payment was received.

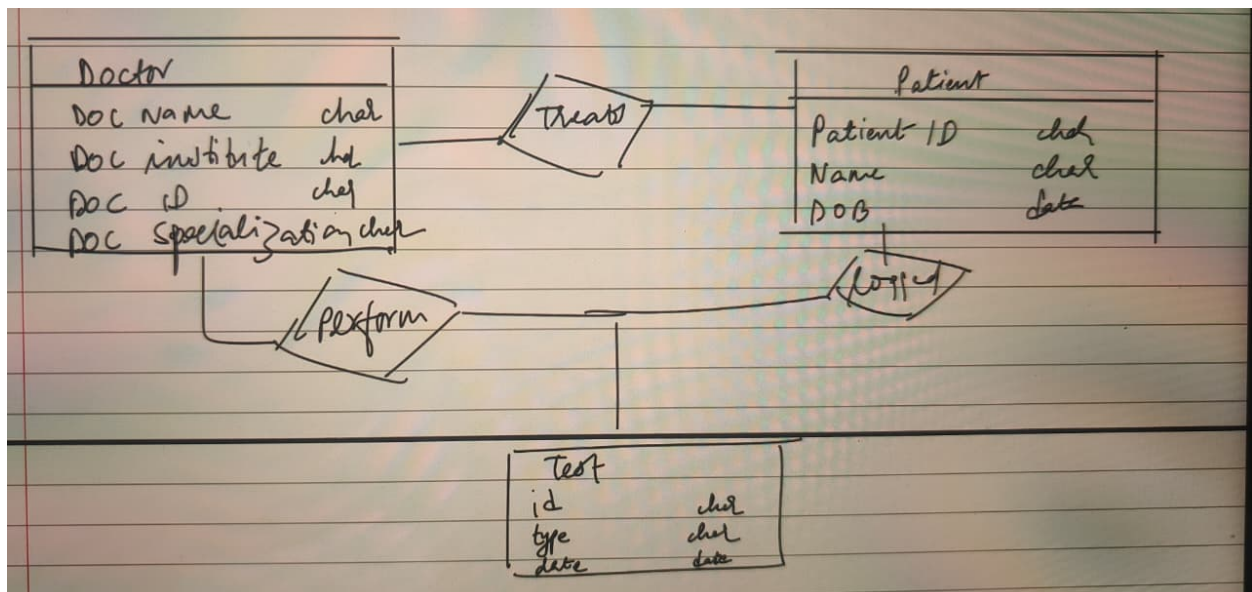
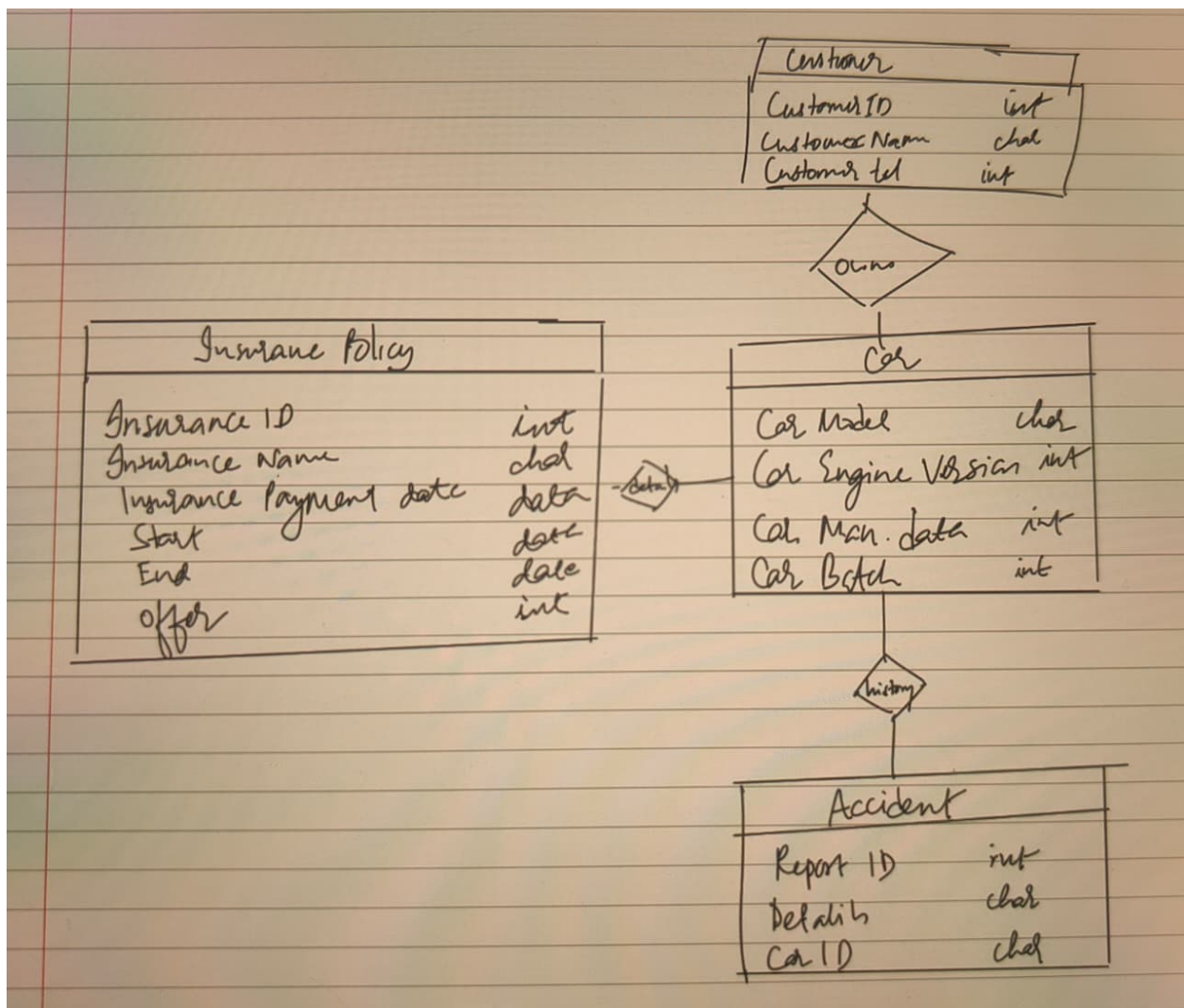


Figure 1: ER Diagram for Car Insurance Company

Q2. Explain the distinctions among the terms primary key, candidate key, and superkey.

- **Superkey:** A set of one or more attributes that, taken collectively, allows us to uniquely identify a tuple in a relation.
- **Candidate Key:** A superkey is minimal; that is, no proper subset of the key is allowed to be a superkey.
- **Primary Key:** A candidate key chosen by the database designer as the principal means of identifying tuples within a relation.

Q3. Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted.



Q4. Explain the difference between a weak and a strong entity set.

Strong Entity Set has a primary key comprising its own attributes. It exists independently in the database schema.

Weak Entity Set does not have sufficient attributes to form a primary key of its own. It is existence-dependent on a strong entity (the identifying owner).

Q5. Consider two entity sets A and B that both have the attribute X (among others whose names are not relevant to this question).

a. **If the two X s are completely unrelated, how should the design be improved?**

They can be renamed to avoid ambiguity in queries and logical representation. For example, rename them to Ax and Bx .

b. **If the two X s represent the same property and it is one that applies both to A and to B , how should the design be improved?**

- **X is the primary key for A but not B :** The design should model a relationship (likely one-to-many) between A and B , where X in B acts as a Foreign Key.
- **X is the primary key for both A and B :** This implies a one-to-one relationship, or that A and B are actually the same entity set. The design should merge them.
- **X is not the primary key for A nor for B :** X should probably be moved to its own table to stop the same data from being repeated in both A and B

Q6. We can convert any weak entity set to a strong entity set by simply adding appropriate attributes (for example, adding a unique ID to every tuple). Why, then, do we have weak entity sets?

Weak entity sets are important for maintaining the logical semantics of the data.

- (a) It clearly shows that one entity's existence depends on another.
- (b) It makes it easier to manage deletions. If a strong entity is removed, the database knows to automatically clear out its dependent weak entities.
- (c) It better reflects how things work in reality. A Course Section doesn't exist in a vacuum; it only exists because a Course exists.

Q7. Explain the distinction between disjoint and overlapping constraints in a generalization/specialization hierarchy.

These constraints determine whether an entity can belong to multiple subclasses simultaneously.

- **Disjoint Constraint:** Specifies that an entity of the superclass can be a member of **at most one** of the subclasses. For example, an *Employee* cannot be both a *Manager* and a *Secretary* at the same time. In an E-R diagram, this is typically noted with a 'd' inside the triangle.

- **Overlapping Constraint:** Specifies that an entity of the superclass can be a member of **more than one** subclass. For example, a *Person* could be both an *Employee* and a *Student* simultaneously. This is noted with an 'o' inside the triangle.

Q8. Explain the distinction between total and partial participation constraints.

Participation constraints specify whether the existence of an entity depends on its being related to another entity via a relationship type.

- **Total Participation:** Every entity in the entity set **must** participate in the relationship.
- **Partial Participation:** Some entities in the entity set may participate in the relationship, but not necessarily all.